

ChargeProof

Trustless cross-chain EV charging settlement.

A station gets paid only when a cryptographically verified charging receipt matches the driver's escrowed intent.



01 · METERED ENERGY

02 · ATTESTED RECEIPT

03 · BOUNDED PAYMENT

EV charging has a settlement gap.

Fragmented records

Drivers, charge-point operators, roaming providers, and payment systems reconcile separate ledgers.

Relayers are assertions

A transaction on chain A cannot natively authorize escrow release on chain B.

Both sides need bounds

Operators need auditable payment. Drivers need capped authorization and automatic refunds.

Shared settlement needs authenticated usage evidence—not another reconciliation file.

Blockchain alone is not cross-chain verification.



Authenticity

Creditcoin needs cryptographic evidence of that exact source transaction.

Semantics

Inclusion alone is insufficient; destination business rules must be revalidated.

Minimized trust

A privileged relayer merely moves the reconciliation trust assumption.

One evidence-bound payment.

01 · INTENT

Bounded escrow

Driver caps spend and fixes station, tariff, expiry, source chain, and registry.



02 · METER

Signed receipt

Authorized device signs deterministic EIP-712 energy and price data.



03 · ANCHOR

Canonical call

Driver finalizes the receipt in the Sepolia source registry.



04 · PROVE

Attestcoin

Inclusion and continuity authenticate the successful transaction.



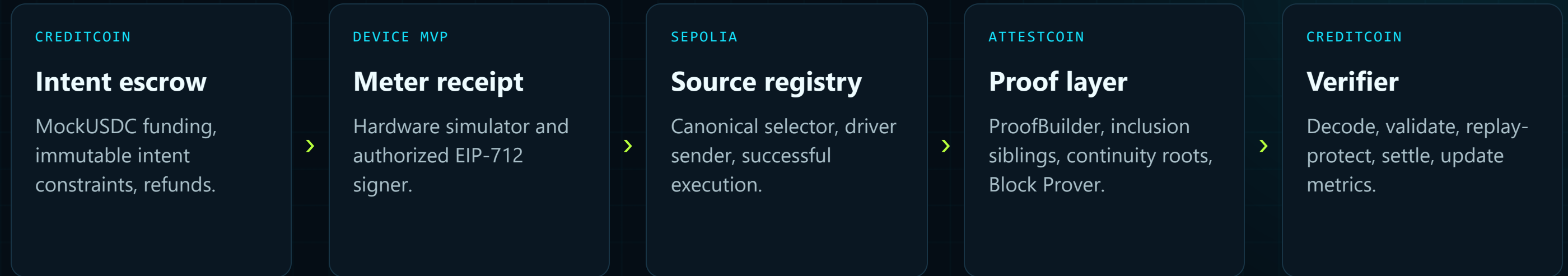
05 · SETTLE

Pay + refund

Operator receives actual cost; unused escrow becomes driver credit.

$$\text{operator payment} = \text{ceil}(\text{energyWh} \times \text{tariff} / 1000) \leq \text{max escrow}$$

Attestcoin closes the verification loop.



Four custom contracts · one resumable worker · one evidence-first dashboard · no bridge custody

Built to be verified, not merely narrated.

GATE 2 COMPLETE · LIVE PROJECT-OWNED EVIDENCE

EVIDENCE	NETWORK	EXPECTED EXPLORER RESULT	STATUS
Funded charging intent	Creditcoin Testnet	0xb9e39a...f871 · 5.00 MockUSDC escrow	Success
Device receipt	Ethereum Sepolia	0x5c7eed...0938 · block 11,539,874	Success
Attestcoin proof	Proof service + Block Prover	2,336 bytes · 7 siblings · 7 roots	0x0FD2 verified
Settlement	Creditcoin Testnet	0xc7ad38...514b · 1.47 / 3.53 MockUSDC	Success
Replay attempt	Creditcoin Testnet	0xb9155e...daff · no accounting change	Mined revert

One real 4,200 Wh receipt crossed Sepolia → Attestcoin → Creditcoin; independent RPC verification confirms settlement and replay protection.

Attestcoin is indispensable.

source chain key	1 · Sepolia
SDK	ProofBuilder.getProof
proof	transaction + siblings + roots
target verifier	native 0x...0FD2
settlement path without proof	impossible

Destination defense in depth

After cryptographic verification, ChargeProof still decodes and checks every business-critical field.

- status
- chain
- source
- selector
- sender
- device
- intent
- tariff
- amount
- expiry
- 3× replay

Remove Attestcoin and Creditcoin has no authenticated Sepolia input—settlement cannot execute.

Trust boundaries are explicit.

BOUNDARY	WHAT CHARGEPROOF GUARANTEES
Physical	Authorized device signature for the MVP—not proof of electricity by Attestcoin.
Source	Expected driver call and successful canonical contract execution.
Cross-chain	Attestcoin transaction inclusion and continuity verification.
Destination	Full semantic validation before escrow state changes.
Payment	Pull credits isolate token transfers; terminal states conserve escrow.

24 automated tests cover tampering, mismatch, invalid proof, expiry, overcharge, replay, refunds, authorization, accounting, and reentrancy.

From EV pilot to a reusable DePIN rail.

NEXT • PROVE

Physical pilot

- Secure-element device keys
- OCPP / OCPI integration
- Independent security audit

THEN • PRODUCTIZE

Operator SDK

- Versioned receipt interface
- Roaming interoperability
- Production payment asset

EXPAND • STANDARDIZE

Metered infrastructure

- Solar microgrids
- Telecom hotspots
- Battery swapping

CEIP candidate: a reusable Creditcoin settlement interface for authenticated external-chain meter receipts.

CLOSE THE LOOP

Metered service → authenticated transaction → bounded payment

Original DePIN vertical slice · Attestcoin at the center · explorer-verifiable · replay-safe

github.com/tang-vu/ChargeProof · BUIDL CTC 2026 Fall